

MINI PROJECT : DATA STRUCTURE AND ALGORITHM

Discuss with your group to propose and apply appropriate data structure concept to complete this mini project. Student can propose their own title (case study) or choose one (1) from the case study below.

Write the title in the google sheet with group information (will be updated).

The case study cannot redundant with other group.

Mini project submission & presentation:

- 1) Documentation
 - i) Front page with title and group information.
 - ii) The synopsis and objective.
 - iii) System requirement, analysis & design (ADT by class diagram)
 - iv) Development activities and coding in C++.
 - v) Task distribution with log sheet that explain individual contribution and group meeting.
- 2) C++ program
 - i) Complete code with comments.
 - ii) Correct – follow the requirements
- 3) Student presentation and demonstration :
 - i) Each group will be given 15 minutes to present and demo.
 - ii) Manage the time so each student in the group can contribute on this session.
- 4) Peer review – will be shared the form

*Note : Student can show the draft before submit the complete version.

* Google sheet to update group and title info : (will be updated)

Case study examples:

- 1) Pizza ordering system using queue
- 2) Car service simulation in C++ using queue
- 3) Virtual Car Promotion and exhibition – simulation in C++ using stack
- 4) Budget Hotel booking using queue
- 5) Vaccine queue simulation system for elderly
- 6) Stack in compiler - Menu driven application in C++
- 7) Phone directory simulation system – simulation in C++ using stack
- 8) Travel car ABC booking system using queue in C++
- 9) Flight and queue simulation at airport
- 10) Swab test vendor machine using stack or queue concept

- 11) Drive thru for vaccine CV19 simulation using queue
- 12) An interview simulation using queue
- 13) Homestay XYZ booking system – queue concept and implementation
- 14) Ayam gunting viral ordering system using queue.
- 15) Book/CD/Car/Maternity set – rental system using queue.
- 16) Stack application – menu driven
- 17) LIFO concept and implementation in visiting the web (a simulation).
- 18) FIFO concept and implementation in graduation (convocation) at the stage (a simulation)
- 19) Text editor simulation using stack
- 20) KoreanFood Queue Simulation Platform Using C++

Submission through e-learning : will be updated

Presentation and demo: will be updated